Homework 6 Solutions

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Section 2.6

Problem 2.6-1

(a) Geometric distribution, $p = 1 - 0.9 = 0.1$. Find $P(X \geq 13)$.

$$\begin{aligned} P(X \geq 13) &= (1-p)^{13-1} \\ &= (0.9)^{12} \\ &\approx 0.2824 \end{aligned}$$

(b) Negative binomial, $r = 3$, $k = 30$, $p = 0.1$.

$$\begin{aligned} P(X = 30) &= \binom{k-1}{r-1} p^r (1-p)^{k-r} \\ &= \binom{29}{2} (0.1)^3 (0.9)^{27} \\ &= 406(0.001)(0.9)^{27} \\ &\approx 0.0236 \end{aligned}$$

Problem 2.6-2

Negative binomial, $r = 5$, $k = 10$, $p = 0.5$.

$$\begin{aligned} P(X = 10) &= \binom{10-1}{5-1} (0.5)^5 (0.5)^{10-5} \\ &= \binom{9}{4} (0.5)^{10} \\ &= 126 \cdot \frac{1}{1024} \\ &= \frac{126}{1024} = \frac{63}{512} \end{aligned}$$

Problem 2.6-4

Geometric distribution, $p = 0.01$. Find $P(X \leq 50)$.

$$\begin{aligned} P(X \leq 50) &= 1 - P(X > 50) \\ &= 1 - (1-p)^{50} \\ &= 1 - (0.99)^{50} \\ &\approx 1 - 0.6050 \\ &= 0.3950 \end{aligned}$$

Section 2.7

Problem 2.7-1

Poisson distribution, $\lambda = 4$.

(a) Find $P(3 \leq X \leq 5)$.

$$\begin{aligned} P(3 \leq X \leq 5) &= P(X=3) + P(X=4) + P(X=5) \\ &= \frac{e^{-4}4^3}{3!} + \frac{e^{-4}4^4}{4!} + \frac{e^{-4}4^5}{5!} \\ &= e^{-4}\left(\frac{64}{6} + \frac{256}{24} + \frac{1024}{120}\right) \\ &\approx 0.5470 \end{aligned}$$

(b) Find $P(X \geq 3)$.

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) = 1 - [P(X=0) + P(X=1) + P(X=2)] \\ &= 1 - e^{-4}\left(\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!}\right) \\ &= 1 - e^{-4}(1+4+8) \\ &= 1 - 13e^{-4} \approx 0.7619 \end{aligned}$$

(c) Find $P(X \leq 3)$.

$$\begin{aligned} P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= e^{-4}\left(\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!}\right) \\ &= e^{-4}\left(1+4+8+\frac{64}{6}\right) \approx 0.4335 \end{aligned}$$

Problem 2.7-5

Poisson distribution.

$$\lambda = \frac{1}{150} \times 225 = 1.5$$

Find $P(X \leq 1)$.

$$\begin{aligned} P(X \leq 1) &= P(X=0) + P(X=1) \\ &= \frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} \\ &= e^{-1.5}(1+1.5) = 2.5e^{-1.5} \\ &\approx 0.5578 \end{aligned}$$

Problem 2.7-8

Poisson approximation to binomial. $n = 1000, p = 0.005$.

$$\lambda = np = 1000 \times 0.005 = 5$$

(a) Find $P(X \leq 1)$.

$$\begin{aligned} P(X \leq 1) &= P(X=0) + P(X=1) \\ &= \frac{e^{-5}5^0}{0!} + \frac{e^{-5}5^1}{1!} \\ &= e^{-5}(1+5) = 6e^{-5} \\ &\approx 0.0404 \end{aligned}$$

(b) Find $P(4 \leq X \leq 6)$.

$$\begin{aligned} P(4 \leq X \leq 6) &= P(X=4) + P(X=5) + P(X=6) \\ &= \frac{e^{-5}5^4}{4!} + \frac{e^{-5}5^5}{5!} + \frac{e^{-5}5^6}{6!} \\ &= e^{-5}\left(\frac{625}{24} + \frac{3125}{120} + \frac{15625}{720}\right) \\ &\approx 0.4971 \end{aligned}$$